

What's New in T&M Tools for Satellite Communication



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Using modular architecture and merging multiple test instruments into one solution with complementing software for test and measurement (T&M) seems to be the trend today. Traditionally, a combination of fixed hardware with an array of resources is employed for testing satellites, and analysing and measuring their signals. But T&M vendors are manufacturing integrated test solutions making it easy for engineers working on satellite communication to upgrade with ease in future. From satellite signal monitoring to signal generation and analysis, there are many equipment from multiple T&M players for enabling establishment of mission-critical satellite systems. In this article, we throw light on some of the latest test equipment for satellite communication and their new capabilities.

One analyser, multiple satellite signals monitoring

In today's world, technology is progressively advancing and data is correspondingly getting complex. Specifically for satellite systems, spot-beam technology makes data more complex as the same frequency is used to transmit various data signals to diverse

geographical locations. One prominent application of spot-beams is in direct broadcast satellite systems used for delivering content via satellites.

Satellite systems must transmit streams of data to and from ground stations via satellite bands, which could comprise various carriers with differing symbol rates and modulation formats. Therefore monitoring of multi-frequency satellite data becomes an extensive measurement challenge. The advent of new high-throughput satellites also brings further measurement challenges as these integrate multiple channels onto a single transponder, which, in turn, results in monitoring of wider bandwidth signals.

To address the challenges and effectively test and analyse satellite signals, Keysight Technologies recently introduced a satellite signal-monitoring reference solution, which primarily provides components of a satellite ground station monitoring system. A satellite ground station monitoring system essentially makes sure satellite spectrum is utilised in an optimal way. In most of the monitoring configurations (sequential scanning and parallel acquisition are two common configurations), a dedicated test set is assigned to a particular frequency or band. For validating satellite-signal integrity, engineers can monitor huge blocks of spectrum and conduct digital-modulation analysis with the help of this solution.

PCI eXtensions for Instrumentation (PXI) is one of the top standards chosen by military and aerospace for high-reliability applications. Based on PXI, this hardware and software solution gives engineers the flexibility to upgrade hardware configurations and use the system for specific signal-monitoring applications such as cross-band timing measurement, carrier power monitoring, logging of measurement results and in-depth modulation analysis for deterministic signals.



Fig. 1: Satellite monitoring using M9393A PXIe VSA with 89600 VSA software (Image courtesy: www.keysight.com)